

Joint Forecasting-and-Judgment RL

Anticipation Task:



Question (q) ⊕ Trajectory Prefix (t) → Guard Model (π_θ)

Sample multiple future summaries

S_1 [Summary bar]

S_2 [Summary bar]

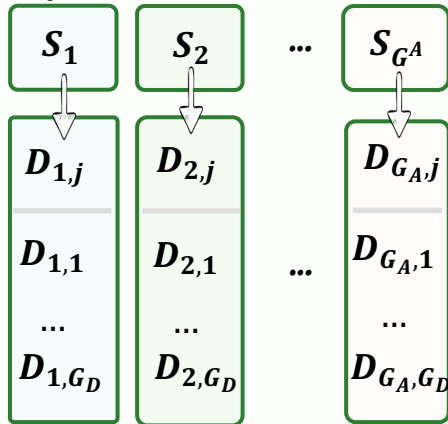
...

S_{G_A} [Summary bar]

Forecasting Reward:

$$r_k^{A, Sim} = Sim(S_k, s)$$

Adjudication Task:



Judgement Reward:

$$r_{k,j}^D = \underbrace{(1 - \lambda_D) l_{k,j}}_{\text{Label correctness}} + \underbrace{\lambda_D \phi_{k,j}}_{\text{Format validity}}$$

Coupled Reward:

Judgement utility for each $r_{k,j}^D$:



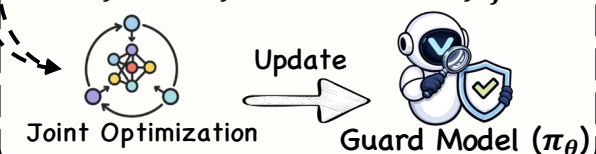
u_k

Forecasting reward balances fidelity and usefulness

$$r_k^A = (1 - \lambda_A) Sim(S_k, s) + \lambda_A u_k$$

$$A_k^A = r_k^A - \text{mean}(\{r_k^A\}_{k=1}^{G_A})$$

$$A_{k,j}^D = r_{k,j}^D - \text{mean}(\{r_{k,j}^D\}_{j=1}^{G_D})$$



Inference:

